Grade 6 Accelerated Name____________________
Unit 15 Date____________________
Lesson 1 Homework Period__________________

For questions 1-4 use the word “**MATHEMATICS**.”

1. What are the possible outcomes of choosing a letter randomly?

2. Which outcomes have more than one possibility?

3. How many outcomes are included in the event that a consonant is chosen?

4. How many outcomes are included in the event that a vowel is chosen?

5. The list of all possible outcomes for an experiment is called the
   - o event.
   - o sample space.
   - o set.
   - o probability.

For questions 6-8, use the standard six-sided dice shown.

6. What is the sample space?

Mike and Meeka are playing a game with the six-sided dice. For each game, determine and explain whether it is fair or not.

7. Mike wins if an odd number is rolled. Meeka wins if an even number is rolled.

   This game is
   - o fair
   - o unfair

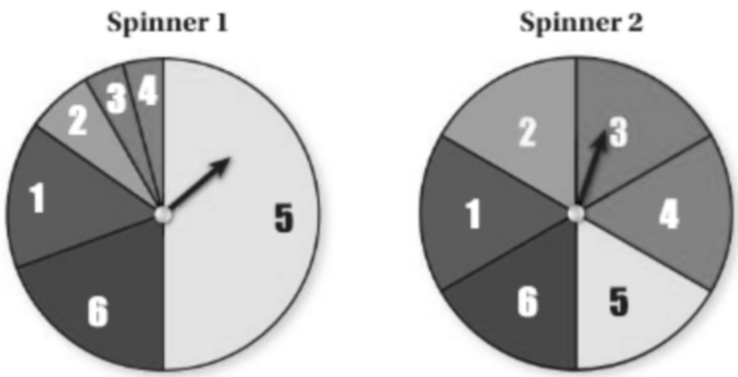
   because ________________________________.

8. Mike wins if a number 4 or less is rolled. Meeka wins if a number 5 or greater is rolled. This game is

o fair
o unfair

because ________________________________.

Use the two spinners shown below for questions 9-10.



9. How many outcomes does each spinner have?

10. One of the spinners is fair and the other is unfair.

Spinner 1 is

o fair
o unfair

because ____________________________________.

Spinner 2 is

o fair
o unfair

because ____________________________________.

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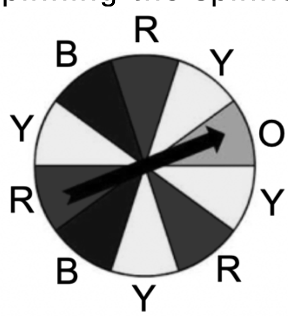
Lesson 2 Homework Period__________________

Simple Experiment	Game	Sample Space
Rolling a fair 6 sided number cube 	Player rolls a standard number cube and wins if it lands on a composite number	
Spinning the spinner 	Player spins the spinner and wins if it lands on a color that is used to make purple.	

1. What are the possible outcomes in the event of rolling a composite number for Game 1?

2. What fraction of the possible outcomes from Game 1 results in a win?

3. What are the possible outcomes in the event of spinning a color that can be used to create purple?

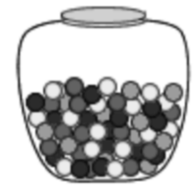
4. What fraction of the possible outcomes from Game 2 results in a win?

5. Which game does a player have a greater likelihood of winning?

For each simple experiment, determine the sample space.

6. A glass jar contains 6 red, 4 green, 3 blue and 5 yellow marbles.

Sample Space



7. The Mathletes club at school must select a day of the school week to meet for their practices.

Sample Space

8. Drawing a face card from a standard deck of cards

Sample Space

9. Flipping a penny

Sample Space

10. Selecting a random month of the year and getting a month that starts with the letter “J”?

Sample Space

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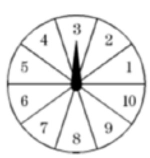
Lesson 3 Homework Period__________________

1. Which of the following values could represent the probability of an event?

A. -1
B. 0.05
C. $\frac{8}{6}$
D. 225%

Determine the probability of each event occurring as a fraction, decimal, and percent. Round decimals to the nearest hundredth if necessary.

	Event	Fraction	Decimal	Percent
2.	Landing on an even number on a 12-sided die  P(even)			
3.	The spinner and lands on a prime number  P(prime)			
4.	Pulling a yellow marble from a jar that has 5 blue, 4 green, and 8 red marbles. P(yellow)			
5.	Picking a day of the week with the letter "n" in the spelling. P(n)			

Use the events from the table for questions 6-7.

6. Explain which event is least likely to occur.

7. Explain which event is most likely to occur.

Samantha is practicing her multiplication facts using the flash cards below. If she chooses a card randomly, find the probability of each even as a fraction, decimal, and percent. Round decimals to the nearest hundredth and percentages rounded to the nearest tenth.

12×3	8×3	6×6	7×4	6×8
7×5	3×12	7×6	8×5	10×6
4×9	12×4	9×6	4×6	8×4

8. $\text{P(product of 36)} =$ ________ = ________ = ________

9. $\text{P(product of 40 or more)} =$ ________ = ________ = ________

10. $\text{P(product } \textit{not} \text{ 48)} =$ ________ = ________ = ________

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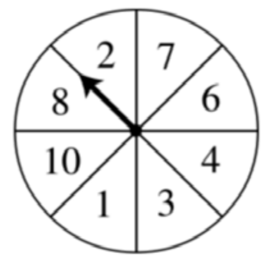
Lesson 4 Homework Period__________________

1. The probabilities of four events are shown. List them in order from least to greatest.

$\frac{4}{11}$ 5 out of 8 55% 0.46

Use the spinner to determine missing events, probabilities, and descriptions of likelihood. Express all probabilities as percentages. Use the vocabulary to complete each sentence.

impossible	unlikely	equally likely as unlikely	likely	certain



2. The probability of landing on an even number, P(even), is _______. This event is _____________.

3. The probability of landing on a number less than 20, P(less than 20), is _______. This event is _____________.

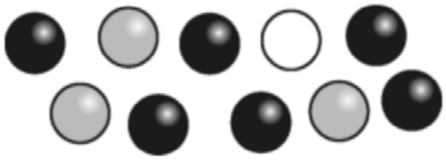
4. The probability of landing on prime number, P(prime), is _______. This event is _____________.

5. The probability of landing on a 5, P(5), is _______. This event is _____________.

6. The probability of landing on a number that is not a 1, P(not a 1), is _______. This event is _____________.

For questions 7-10, determine which event is more likely.

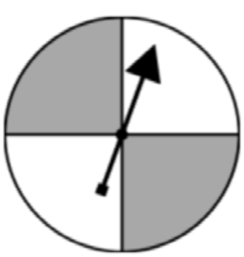
7. Martin put 10 marbles in a jar. The probability he chooses a black marble is 0.6. The probability of choosing a white marble is 10%. Which is more likely to occur?



8. Two students roll the die. Player A wins if they roll an even number. Player B wins if they roll a number less than 3. Who has the higher probability of winning? Explain.

9. Fort Lauderdale airport reported that on a given day, the probability that an airline arrived on time was 0.435. The likelihood that the next airplane arrived earlier than scheduled was $\frac{1}{6}$ and the later than scheduled was 27%. What is most likely to occur when the next airplane arrives?

10. The following spinner is spun. On the next spin, which color is the spinner more likely to land on? Explain.



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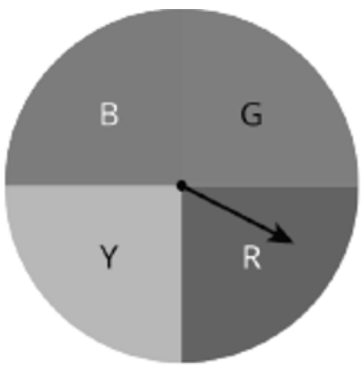
Lesson 5 Homework Period__________________

1. Explain the difference between theoretical and experimental probability of a simple experiment.

Tyriek performed a simulation and spun the arrow on the spinner below. He recorded his results in the frequency table below.



Color	**Frequency**
Blue	7
Green	9
Red	4
Yellow	5

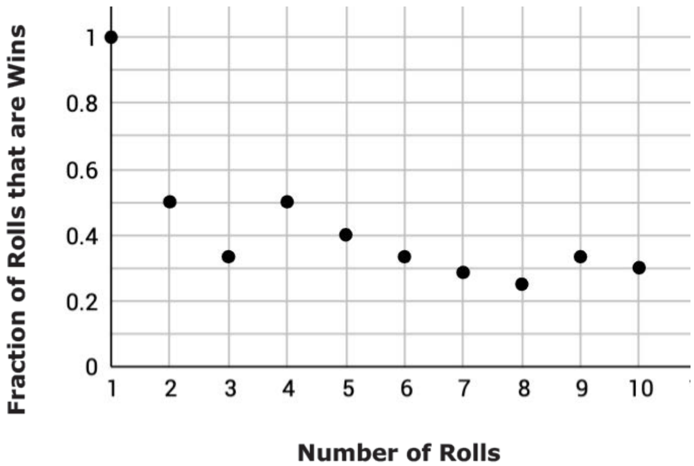
2. What is the theoretical probability of landing on red?

3. What is the experimental probability of landing on red?

Arleni is playing a game where she wins if she rolls a 5 or a 6 on a standard number cube.

4. What is the theoretical probability that Alreni wins the game?

She then performs the simulation 10 times. The graph below shows the fraction of games that are wins.



5. Estimate the probability of a roll winning this game based on the graph.

6. What type of probability is this?

7. Draw a horizontal line on the graph representing the theoretical probability of rolling a 5 or a 6.

8. After 10 rolls, what fraction of the total rolls were a win?

9. If Arleni rolled the number cube 10 more times, estimate the probability that she will roll a 5 or a 6.

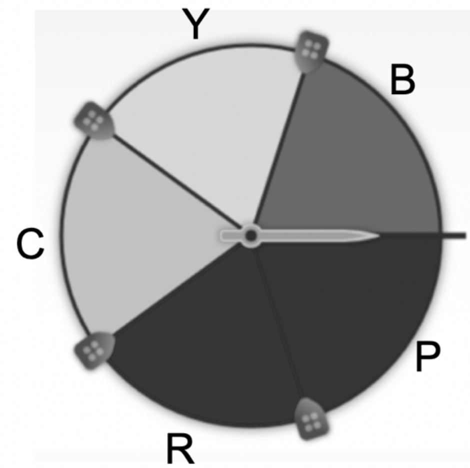
10. If she were to roll the number cube 500, how many times would you expect for her to roll a 5 or a 6

Grade 6 Accelerated
Unit 15
Lesson 6 Homework

Name____________________
Date_____________________
Period___________________

1. Find the theoretical probability of landing on each color as a percentage rounded to the nearest tenth.

Color	Theoretical Probability
Cyan	
Yellow	
Blue	
Purple	
Red	



A computer simulator spinner performed more trials. The results as percentages are shown in the table below.

Color	Simulation 1 (10 trials)	Simulation 2 (100 trials)	Simulation 3 (1,000 trials)
Cyan	0%	19%	
Yellow	30%	23%	
Blue	30%	21%	
Purple	10%	16%	
Red	30%	21%	

2. What do you notice in comparing Simulation 1 to Simulation 2?

3. Complete the table by estimating the percentages for Simulation 3.

Your teacher gives you a standard number cube with numbers 1–6 on its faces. You roll it 500 times. Your results are shown in the table below.

Outcome	1	2	3	4	5	6
Frequency	78	92	75	90	76	89

4. What is the experimental probability of rolling an even number?

5. How many times would you expect to roll an even number?

6. As the number of trials in this simple experiment increases, the
   - o experimental probability
   - o theoretical probability

   gets closer to the
   - o experimental probability.
   - o theoretical probability.

Four situations are described. For each situation, create a possible simulation to fairly represent it.

7. Six people are going out to lunch together. One of them will be selected at random to choose which restaurant to go to. Who gets to choose?

8. When a toddler walks, it is equally likely to step forward with its left foot or its right foot. Which foot will it use for its first step?

9. Your school is taking 4 buses of students on a field trip. Will you be assigned to the same bus that your math teacher is riding on?

10. An experiment will produce one of ten different outcomes with equal probability for each. Why would using a standard number cube to simulate the experiment be a bad choice?

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Lesson 7 Homework Period__________________

The Smith family has 3 children and want one more child to complete their family. $\frac{1}{3}$ of the children are girls and the rest are boys.

1. What is the probability that the next child will be a girl, represented as a percent rounded to the nearest hundredth?

2. What is the theoretical probability that the next child will be a girl?

Describe how each of the following can be used to simulate if the next child will be a girl.

3. A spinner with 10 equal sections.

4. A fair six-sided die.

5. A standard 52-card deck.

The weather forecast says there is a 50% chance it will rain today.

Describe how each of the following can be used to simulate if it will rain today.

6. A coin

7. Different numbers in a box

There are 4 types of blood: A, B, AB, and O. Blood type is determined by genes inherited from parents. Suppose that a type AB blood donor is needed at a local hospital for a patient. 0.4 donors have type AB blood.

8. Find the probability that the next donor will not have type AB blood?

9. What is the theoretical probability the next donor will have type AB blood?

10. Describe a way to simulate the probability the next donor will have type AB blood?